

The Asymptotic Distribution of the Order of Elements in Symmetric Semigroups*

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Communicated by Managing Editors

Received January 1, 1972

P. Erdős and P. Turán [8] (*Acta Math. Acad. Sci. Hungar.*, **18** (1967), 309–320) have shown that, if $K(n, x)$ is the number of elements P in S_n , the symmetric group on n letters, whose order $O(P)$ satisfies

$$\log O(P) \leq \frac{1}{2} \log^2 n + (1/\sqrt{3}) x \log^{3/2} n,$$

then

$$\lim_{n \rightarrow \infty} K(n, x)/n! = (\sqrt{2\pi})^{-1} \int_{-\infty}^x e^{-t^2/2} dt.$$

In this paper the analogous result for the symmetric semigroup is obtained. Let $\alpha \in T_n$, the symmetric semigroup on n letters (the set of all mappings of $\{1, 2, \dots, n\}$ into $\{1, 2, \dots, n\}$) and let $O(\alpha)$ be the order of α . If $L(n, x)$ is the number of $\alpha \in T_n$ with

$$\log O(\alpha) \leq \frac{1}{8} \log^2 n + \left(\frac{1}{\sqrt{24}}\right) x \log^{3/2} n,$$

then

$$\lim_{n \rightarrow \infty} L(n, x)/n^n = (\sqrt{2\pi})^{-1} \int_{-\infty}^x e^{-t^2/2} dt.$$

1. INTRODUCTION

In the papers by P. Erdős and P. Turán [6, 7, 8, 9], various questions concerning statistical characteristics of the symmetric group on n letters, S_n , for large n are studied. In particular, let P denote a generic element of

* Sponsored in part by The United States Army under Contract DA-31-124-ARO-D-462, the Netherlands Ministry of Education, Science, and Art, and the Bolyai János Matematikai Társulat, Budapest, Hungary.

S_n and let $O(P)$ be the order of P . Then, if $K_{n,x}$ is the number of elements of S_n satisfying

$$\log O(P) \leq \frac{1}{2} \log^2 n + x \log^{3/2} n, \quad (1)$$

we have

$$\lim_{n \rightarrow \infty} \frac{K(n, x)}{n!} = \sqrt{\frac{3}{2\pi}} \int_{-\infty}^x e^{-3t^2/2} dt. \quad (2)$$

Equivalently, if P is "chosen at random" from S_n , define

$$Z = \left(\log O(P) - \frac{1}{2} \log^2 n \right) / \frac{1}{\sqrt{3}} \log^{3/2} n;$$

then the distribution of Z converges (as $n \rightarrow \infty$) to the standard normal distribution.

In this note the analogous result to (1) and (2) for the symmetric semigroup is obtained.

In order to introduce this theorem some definitions and notation are required. The symmetric semigroup T_n is the set of all mappings of $X_n = \{x_1, x_2, \dots, x_n\}$ into X_n . With no loss of generality, we can take $X_n = \{1, 2, \dots, n\}$. If $\alpha, \beta \in T_n$, then the product $\alpha\beta \in T_n$ is defined by $(\alpha\beta)(x) = \alpha(\beta(x))$, for all $x \in X_n$. There is a one-to-one correspondence between elements of T_n and the class of labeled directed graphs on n vertices, with each vertex having exactly one edge departing from it. Such a graph may be constructed by the following procedure: if $\alpha(x) = x_1$, draw the directed edge from x to x_1 (if $x = x_1$, draw a loop at x_1). For each α we divide X_n into two classes, cyclical and non-cyclical elements. An $x \in X_n$ is said to be cyclical under α if there is an $m > 0$ with $\alpha^m(x) = x$. The cyclical elements under α will be denoted by C_α . The restriction of α to C_α will be denoted by α^* . α^* is a one-to-one mapping of C_α onto C_α and thus permutes the elements of C_α . Dénes [1, 2, 3, 4, 5] has called α^* the main permutation of α . We let $L = |C_\alpha|$. Also, for each $x \in X_n$, there is a least integer $r = r_\alpha(x) \geq 0$ such that $\alpha^r(x) \in C_\alpha$. r is called the α -height of x . The height of α is defined by $h(\alpha) = \max_{x \in X_n} r_\alpha(x)$.

Then, by analogy with the definition of $O(P)$, $P \in S_n$, Dénes [1, 3, 5] defined the order of α , $O(\alpha)$, $\alpha \in T_n$ as the number of distinct elements of T_n in the set $\{\alpha, \alpha^2, \dots\}$. He has shown that this is equivalent to $O(\alpha) = O(\alpha^*) + \max(0, h(\alpha) - 1)$; and this is also equivalent to defining $O(\alpha)$ as the least m such that, for $0 < q \leq m$, $\alpha^q = \alpha^{m+1}$.

2. THE ASYMPTOTIC DISTRIBUTION OF $O(\alpha)$

We now obtain the analog of (1) and (2). We will use language and notation equivalent to assuming that the n^n elements of T_n are chosen at random and equally likely. This entails no loss of generality for our purposes. Then α will be a random variable taking values in T_n . We first produce some preliminary lemmas that will be subsequently employed in the proof of the theorem.

LEMMA 1. Let $k_1(n)$ and $k_2(n)$ be any two functions with

$$\lim_{n \rightarrow \infty} k_1(n)/n^{1/2} = 0 \quad \text{and} \quad \lim_{n \rightarrow \infty} k_2(n)/n^{1/2} = \infty,$$

then

$$\lim_{n \rightarrow \infty} P\{k_1(n) < L < k_2(n)\} = 1. \quad (3)$$

Proof. From (3.12) of B. Harris [10], we have

$$P\{L = l\} = \frac{(n-1)!}{(n-l)!} \frac{l}{n^l}, \quad l = 1, 2, \dots, n,$$

and hence

$$P\{L \leq l\} = \sum_{j=1}^l \frac{(n-1)!}{(n-j)!} \frac{j}{n^j} \leq \frac{1}{n} \sum_{j=1}^l j \leq \frac{l^2}{n}. \quad (4)$$

Also

$$\begin{aligned} P\{L \geq m\} &= \sum_{j=m}^n \frac{(n-1)!}{(n-j)!} \frac{j}{n^j} \\ &= \sum_{j=m}^n \frac{j}{n} \prod_{i=1}^{j-1} \left(1 - \frac{i}{n}\right) \\ &\leq \sum_{j=m}^n \frac{j}{n} e^{-j(j-1)/2n}. \end{aligned}$$

Now, for $m \geq \sqrt{n} + 1$, the terms in the sum are decreasing and $j(j-1) \geq j^2/2$ for $j \geq 2$. Hence, for $m \geq \max(3, \sqrt{n} + 1)$,

$$\begin{aligned} \sum_{j=m}^n \frac{j}{n} e^{-j(j-1)/2n} &< \int_{m-1}^n \frac{x}{n} e^{-x(x-1)/2n} dx \\ &< \int_{m-1}^n \frac{x}{n} e^{-x^2/4n} dx \\ &\leq 2e^{-(m-1)^2/4n}. \end{aligned}$$

Thus we have shown that

$$P\{k_1(n) < L < k_2(n)\} > 1 - \frac{k_1^2}{n} - 2e^{-(k_2-1)^2/4n}$$

and hence tends to unity as $n \rightarrow \infty$.

We now establish the following lemma:

LEMMA 2. Let $\gamma > 0$ be arbitrarily small and let $l = l(n)$ satisfy

$$(\tfrac{1}{2} - \gamma) \log^2 l - \log n \rightarrow \infty \quad (5)$$

as $n \rightarrow \infty$. Then

$$\begin{aligned} \lim_{n \rightarrow \infty} P\{\sqrt{3} \log^{-3/2} l (\log O(\alpha) - \tfrac{1}{2} \log^2 l) \leq x \mid L = l\} \\ = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt. \end{aligned} \quad (6)$$

Proof. By definition

$$O(\alpha) = O(\alpha^*) + \max(0, h(\alpha) - 1).$$

Hence, for $|C_\alpha| = l$,

$$O(\alpha^*) \leq O(\alpha) \leq O(\alpha^*) + n - l - 1. \quad (7)$$

Let E be the event $C_\alpha = \{x_{i1}, x_{i2}, \dots, x_{il}\}$, where the x_{ij} 's are l distinct elements of X_n . Then it is easily seen that

$$P\{\alpha^* = \alpha_0^* \mid E\} = 1/l!,$$

where α_0^* is any specified permutation of $\{x_{i1}, x_{i2}, \dots, x_{il}\}$. From this it follows that conditional on the event $L = l$, the results (1) and (2) of Erdős and Turán apply to the distribution of $O(\alpha^*)$.

Also, from (7), for all $\alpha \in T_n$,

$$\begin{aligned} \log O(\alpha^*) &\leq \log O(\alpha) \leq \log \left\{ O(\alpha^*) \left(1 + \frac{n}{O(\alpha^*)} \right) \right\} \\ &\leq \log O(\alpha^*) + \frac{n}{O(\alpha^*)}. \end{aligned} \quad (8)$$

Thus

$$\begin{aligned} P \left\{ \sqrt{3} \log^{-3/2} l \left(\log O(\alpha^*) + \frac{n}{O(\alpha^*)} - \frac{1}{2} \log^2 l \right) \leq x \mid L = l \right\} \\ \leq P\{\sqrt{3} \log^{-3/2} l (\log O(\alpha) - \tfrac{1}{2} \log^2 l) \leq x \mid L = l\} \\ \leq P\{\sqrt{3} \log^{-3/2} l (\log O(\alpha^*) - \tfrac{1}{2} \log^2 l) \leq x \mid L = l\}. \end{aligned} \quad (9)$$

For any $\gamma > 0$, let E_γ be the event

$$O(\alpha^*) \geq e^{(\frac{1}{2}-\gamma)\log^2 l}. \quad (10)$$

Then

$$\begin{aligned} & P \left\{ \sqrt{3} \log^{-3/2} l \left(\log O(\alpha^*) + \frac{n}{O(\alpha^*)} - \frac{1}{2} \log^2 l \right) \leq x \mid L = l \right\} \\ &= P \left\{ \sqrt{3} \log^{-3/2} l \left(\log O(\alpha^*) + \frac{n}{O(\alpha^*)} - \frac{1}{2} \log^2 l \right) \leq x, E_\gamma \mid L = l \right\} \\ &+ P \left\{ \sqrt{3} \log^{-3/2} l \left(\log O(\alpha^*) + \frac{n}{O(\alpha^*)} - \frac{1}{2} \log^2 l \right) \leq x, E_\gamma^c \mid L = l \right\}. \end{aligned} \quad (11)$$

Choose any $\delta > 0$. Then, from Erdős and Turán [6], there exists an $l_0(\delta, \gamma)$ such that

$$P\{E_\gamma \mid L = l\} \geq 1 - \delta, \quad (12)$$

for every $l > l_0$. Further, from (1) and (2), there is an $l_1(\delta)$ such that, for all $l > l_1$,

$$\left| P\left\{ \sqrt{3} \log^{-3/2} l \left(\log O(\alpha^*) - \frac{1}{2} \log^2 l \right) \leq x \mid L = l \right\} - \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt \right| \leq \delta. \quad (13)$$

Thus, for $l > l_0$, there is a $K(l, \gamma)$, with $|K(l, \gamma)| < \delta$ such that from (10) and (12), we get

$$\begin{aligned} & P \left\{ \sqrt{3} \log^{-3/2} l \left(\log O(\alpha^*) + \frac{n}{O(\alpha^*)} - \frac{1}{2} \log^2 l \right) \leq x \mid L = l \right\} \\ &= P \left\{ \sqrt{3} \log^{-3/2} l \left(\log O(\alpha^*) + \frac{n}{O(\alpha^*)} - \frac{1}{2} \log^2 l \right) \leq x, E_\gamma \mid L = l \right\} \\ &+ K(l, \gamma) \\ &\geq P \left\{ \sqrt{3} \log^{-3/2} l \left(\log O(\alpha^*) + e^{\log n - (\frac{1}{2}-\gamma)\log^2 l} - \frac{1}{2} \log^2 l \right) \leq x \mid L = l \right\} \\ &+ K(l, \gamma). \end{aligned} \quad (14)$$

From (5), there exists an n_0 such that, for $n > n_0$, there is an $l_2(n, \delta, \gamma)$ with

$$e^{\log n - (\frac{1}{2}-\gamma)\log^2 l} < \delta, \quad (15)$$

for all $l > l_2$.

Hence, from (14) and (15),

$$\begin{aligned} P \left\{ \sqrt{3} \log^{-3/2} l \left(\log O(\alpha^*) + \frac{n}{O(\alpha^*)} - \frac{1}{2} \log^2 l \right) \leq x \mid L = l \right\} \\ \geq P \{ \sqrt{3} \log^{-3/2} l (\log O(\alpha^*) - \frac{1}{2} \log^2 l) \leq x - \delta \mid L = l \} + K(l, \gamma). \end{aligned} \quad (16)$$

Thus, for $l > \max(l_0, l_1, l_2)$, combining (10), (13), and (16), we get

$$\begin{aligned} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x-\delta} e^{-t^2/2} dt - \delta \\ \leq P \{ \sqrt{3} \log^{-3/2} l (\log O(\alpha) - \frac{1}{2} \log^2 l) \leq x \mid L = l \} \\ \leq \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt + \delta. \end{aligned}$$

Since (5) implies $l \rightarrow \infty$ as $n \rightarrow \infty$, the lemma is established.

We now establish the theorem:

THEOREM. *Let $\alpha \in T_n$, the symmetric semigroup on n letters. Then, the asymptotic ($n \rightarrow \infty$) distribution of the order of α , $O(\alpha)$ is given by*

$$\begin{aligned} \lim_{n \rightarrow \infty} P \{ \sqrt{24} \log^{-3/2} n (\log O(\alpha) - \frac{1}{8} \log^2 n) \leq x \} \\ = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt. \end{aligned} \quad (17)$$

Proof. Let

$$A_n = \left\{ \alpha \in T_n : \frac{n^{1/2}}{\log n} < l < n^{1/2} \log n \right\}.$$

Then, for $n = 1, 2, \dots$, $-\infty < x < \infty$, let $u = u(n, x)$ be given and

$$\begin{aligned} P \{ \log O(\alpha) \leq u(n, x) \} &= \sum_{l \in A_n} P \{ L = l \} P \{ \log O(\alpha) \leq u \mid L = l \} \\ &\quad + \sum_{l \in A_n^c} P \{ L = l \} P \{ \log O(\alpha) \leq u \mid L = l \}. \end{aligned} \quad (18)$$

Now

$$\begin{aligned} \sum_{l \in A_n^c} P \{ L = l \} P \{ \log O(\alpha) \leq u \mid L = l \} \\ \leq \sum_{l \in A_n^c} P \{ L = l \} = P \{ A_n^c \}. \end{aligned}$$

By Lemma 1, for any $\delta > 0$, there is an n_0 such that, for all $n > n_0$, $P\{A_n^c\} < \delta$. Further, there is an n_1 such that, for all $n > n_1$, and all $l \in A_n$, $\log^2 l \geq \frac{1}{8} \log^2 n$. Thus the hypotheses of Lemma 2 are then satisfied. Hence, for any $\eta > 0$, there is an n_2 so that, for all $n > n_2$, and for every $l \in A_n$, we have

$$\left| P \left\{ \log O(\alpha) \leq \frac{1}{2} \log^2 l + \frac{x}{\sqrt{3}} \log^{3/2} l \mid L = l \right\} - \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt \right| < \eta. \quad (19)$$

It can easily be seen that $l \in A_n$ implies

$$\frac{1}{2} \log^2 l + \frac{x}{\sqrt{3}} \log^{3/2} l = \frac{1}{8} \log^2 n + q(l) + \frac{x}{\sqrt{24}} \log^{3/2} n(1 + r(l)), \quad (20)$$

where

$$|q(l)| \leq 2 \log n \log \log n$$

and

$$|r(l)| \leq 4 \log \log n / \log n$$

for all $n > n_3$.

Hence, for every $l \in A_n$, from (20)

$$\begin{aligned} P\{\sqrt{3} \log^{-3/2} l (\log O(\alpha) - \frac{1}{2} \log^2 l) \leq x \mid L = l\} \\ &= P\{\sqrt{24} \log^{-3/2} n(1 + r(l))^{-1} (\log O(\alpha) - \frac{1}{8} \log^2 n - q(l)) \leq x \mid L = l\} \\ &= P\{\sqrt{24} \log^{-3/2} n (\log O(\alpha) - \frac{1}{8} \log^2 n - q(l)) \leq x(1 + r(l)) \mid L = l\} \\ &= P\{\sqrt{24} \log^{-3/2} n (\log O(\alpha) - \frac{1}{8} \log^2 n) \\ &\leq x(1 + r(l)) + \sqrt{24} q(l) \log^{-3/2} n \mid L = l\}. \end{aligned} \quad (21)$$

Thus for $M > 0$, all $|x| \leq M$ and any $\epsilon > 0$, for $n > n_4$ (19) and (21) show that

$$\begin{aligned} &\left| P\{\sqrt{24} \log^{-3/2} n (\log O(\alpha) - \frac{1}{8} \log^2 n) \right. \\ &\quad \left. \leq x + \rho(x) \mid L = l\} - \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt \right| < \eta, \end{aligned} \quad (22)$$

where $|\rho(x)| < \epsilon$ for all $|x| \leq M$.

$$\begin{aligned} &\left| P\{\sqrt{24} \log^{-3/2} n (\log O(\alpha) - \frac{1}{8} \log^2 n) \right. \\ &\quad \left. \leq x \mid L = l\} - \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt \right| < 2\eta \end{aligned} \quad (23)$$

for all $|x| < M$, $n > \max_{1 \leq i \leq 4} n_i$, and all $l \in A_n$.

Thus, for $|x| < M$ and $n > \max_{0 \leq i \leq 4} n_i$,

$$\begin{aligned} P \left\{ \log O(\alpha) \leq \frac{1}{8} \log^2 n + \frac{x}{\sqrt{24}} \log^{3/2} n \right\} \\ \leq \sum_{l=1}^n P\{L = l\} \left(\frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt + 2\eta \right) + \delta \\ = \left(\frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt + 2\eta \right) \sum_{l=1}^n P\{L = l\} + \delta \\ \leq \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt + 2\eta + \delta. \end{aligned}$$

Similarly,

$$\begin{aligned} P \left\{ \log O(\alpha) \leq \frac{1}{8} \log^2 n + \frac{x}{\sqrt{24}} \log^{3/2} n \right\} \\ \geq \sum_{l=1}^n P\{L = l\} \left(\frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt - 2\eta \right) - \delta - 2\eta\delta. \end{aligned}$$

Since, for any $\epsilon > 0$, M may be chosen so that

$$\frac{1}{\sqrt{2\pi}} \int_{-M}^M e^{-t^2/2} dt > 1 - \epsilon,$$

the theorem is established.

ACKNOWLEDGMENTS

The author wishes to thank Dr. Jozsef Dénes and Professor Paul Erdős for bringing this problem to his attention during conversations while the author was visiting Budapest.

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